1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading, and time synchronization errors, it is essential to establish a baseline model of Over-the-Air (OTA) computation under ideal conditions. The foundational premise of OTA computing is its ability to reduce the communication latency of data aggregation from $\mathcal{O}(K)$ to $\mathcal{O}(1)$ by allowing K devices to transmit simultaneously [3].

In this idealized scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted solely by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, algorithmic MATLAB modeling, block-level Simulink architecture, and GNU Radio flowgraphs that demonstrate this ideal, unimpaired baseline. Establishing this benchmark is critical for quantifying the exact degradation caused by physical channel constraints in subsequent chapters.

1.a Mathematical Formulation of the Ideal Channel

Consider a distributed wireless sensor network consisting of K End Device (ED)s transmitting local analog values to a central Fusion Center (FC). Let $x_k(t)$ denote the continuous-time, real-valued signal transmitted by the k-th device. For analog OTA computation, uncoded Pulse Amplitude Modulation (PAM) is widely considered the optimal transmission scheme, as it linearly maps the sensor readings directly to the amplitude of the transmitted waveform [2], [4].

In an ideal Multiple Access Channel (MAC), we assume perfect Channel State Information (CSI) at the transmitters, which allows for flawless pre-equalization of phase shifts and path loss attenuation. Consequently, the electromagnetic waves superimpose perfectly at the receiving antenna of the FC. The received baseband signal $y(t)$ is mathematically expressed as the linear sum of the transmitted signals plus noise:

$$y(t) = \sum_{k=1}^{K} x_k(t) + n(t) \tag{1}$$

where $n(t) \sim \mathcal{N}(0, \sigma^2)$ represents the AWGN with zero mean and variance σ^2. Because there is no multipath delay spread in this baseline model, the channel impulse response is effectively a Dirac delta function $h(t) = \delta(t)$. This implies that the current symbol is entirely unaffected by past or future symbols, resulting in zero ISI. To extract the arithmetic average of the K sensor readings, the FC simply applies a post-processing scaling factor of $1/K$ to the received aggregated signal $y(t)$.

1.b MATLAB Simulation and Code Analysis

To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

```
1
```

```
        % The Air acts as an ideal adder (MAC Superposition)
        Ideal_Sum = sum(X, 1);

        % Add AWGN
        Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');

        % Calculate the Fusion Center Average
        Estimated_Average = Received_Signal / K;

        % Display Mean Squared Error (MSE)
        MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE
           at %d dB SNR: %f\n', SNR_dB, MSE);
```

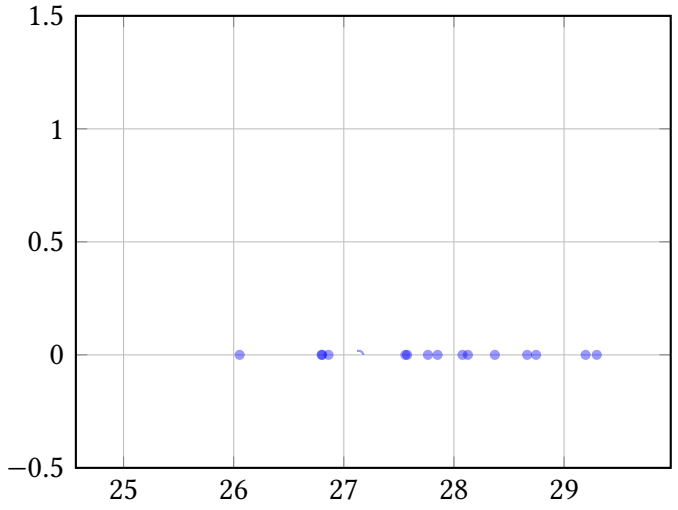
While Listing 1 demonstrates the fundamental principle in its unadulterated form, practical wireless systems necessitate pulse shaping to satisfy spectral constraints and facilitate signal recovery. Consequently, a more physically realistic model incorporates matched Raised Cosine (RC) filtering at both the transmitter and receiver. This architectural refinement introduces additional degrees of freedom—namely, the Symbols per Sample (sps) parameter and the roll-off factor (α)—which materially influence system performance. The enhanced implementation is presented in Listing 2.

Listing 2: MATLAB script for OTA computation aggregation using the Root Raised Cosine (RRC) filter.

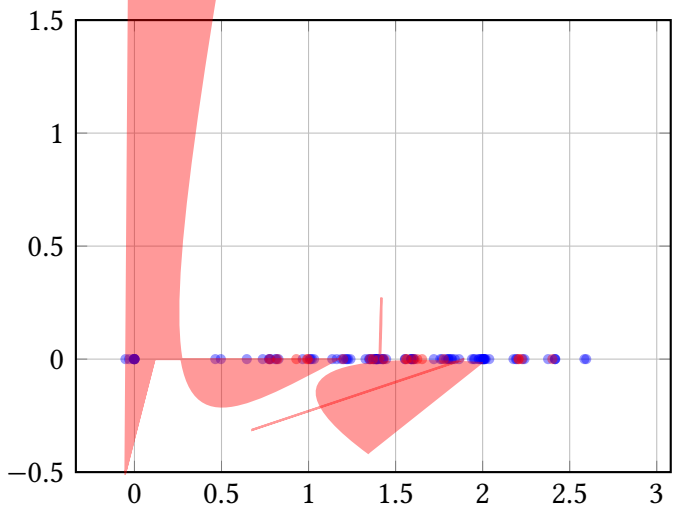
```
        txUpSamped = upsample(X', sps_val); % phase may need to be set

        % Raised Cosine Filter
        rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");

        rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);

        for k = 1:K
            rx_agg_signal = rx_agg_signal + conv(txUpSamped(:,k), rcFilter(:));
        end

        Received_Signal = awgn(rx_agg_signal, current_snr, 'measured');

        matchedSignal = conv(Received_Signal, flipud(rcFilter(:)), 'same');

        rxMatched = downsample(matchedSignal, sps_val);

        start_idx = floor(span/2); stop_idx = floor(span/2) + mod(span,2) + 1;

        % Truncate edges to avoid filter transient errors
        Estimated_Average = rxMatched(start_idx:end-stop_idx, :) / K;

        % Calculate and store MSE in the matrix
        mse_results(snr_idx, a_idx) = immse(Estimated_Average', mean(X, 1));
```

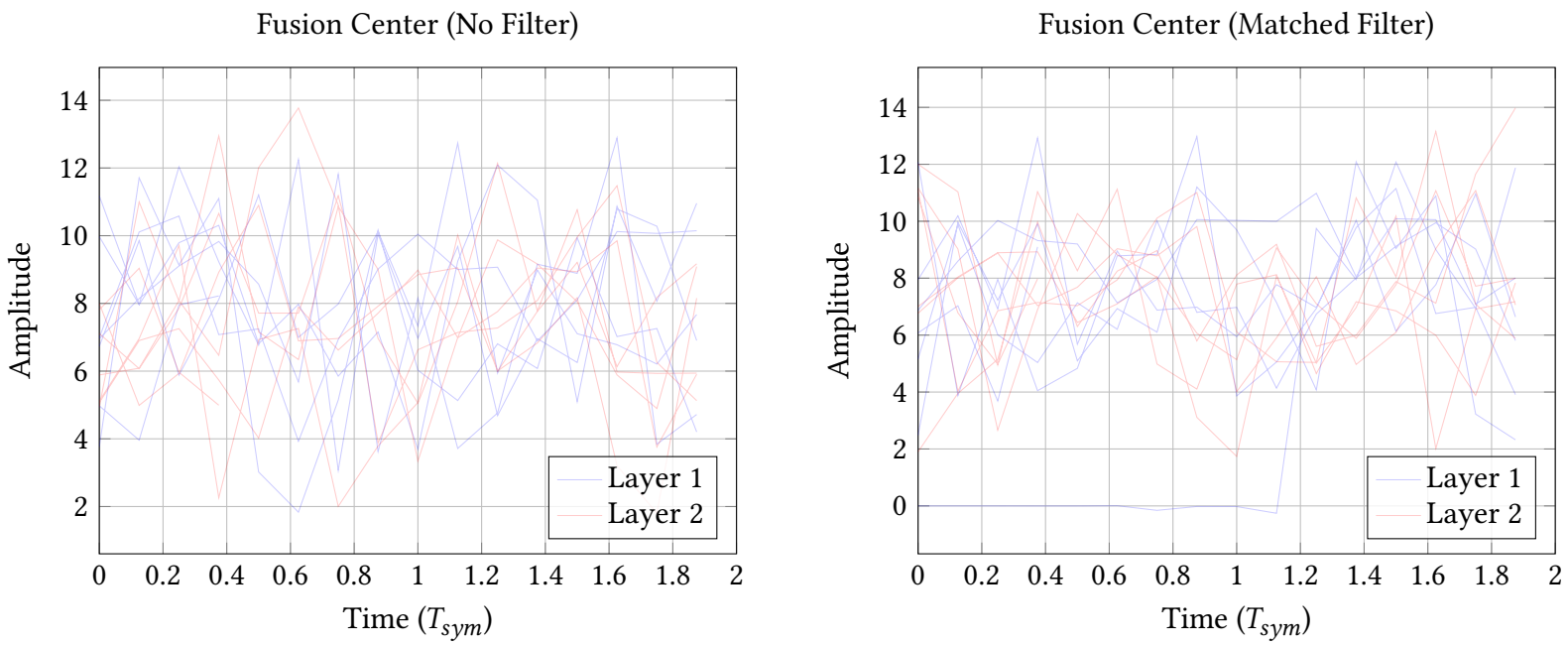
Computational execution of the aforementioned algorithms reveals a critical distinction between the idealized and practical

stages, enabling quantitative assessment of performance degradation attributable to finite symbol-rate sampling and spectral roll-off constraints.



The plot corresponding to the fusion center observation after the matched RRC filter shows a more compact and well-defined symbol distribution, indicating that the filtering stage has improved temporal alignment and reduced the impact of inter-symbol interference. In contrast, the constellation associated with the theoretical receiver model without pulse shaping appears more dispersed and less structured, reflecting the absence of the smoothing and energy-concentrating effect provided by the matched filter. This comparison highlights that the matched RRC filter contributes not only to spectral shaping, but also to preserving symbol integrity and improving the reliability of the recovered OTA estimate.





(a) Unfiltered signal.

(b) Signal after RRC filtering.

Figure 8: Comparative eye diagrams demonstrating the signal reconstruction at the fusion center.

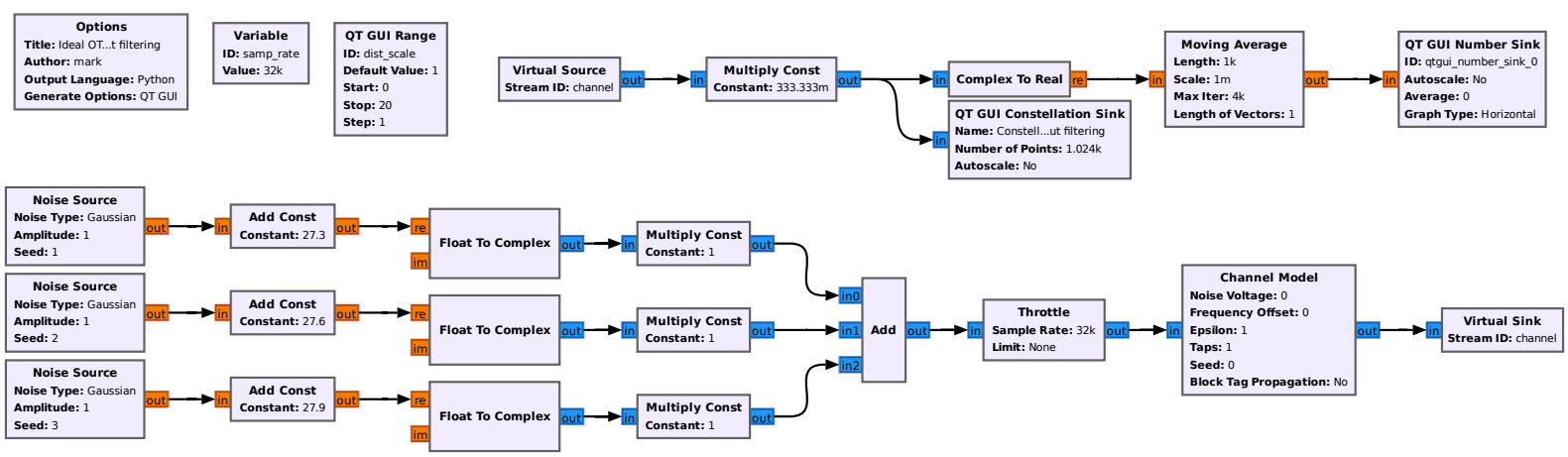


Figure 9: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

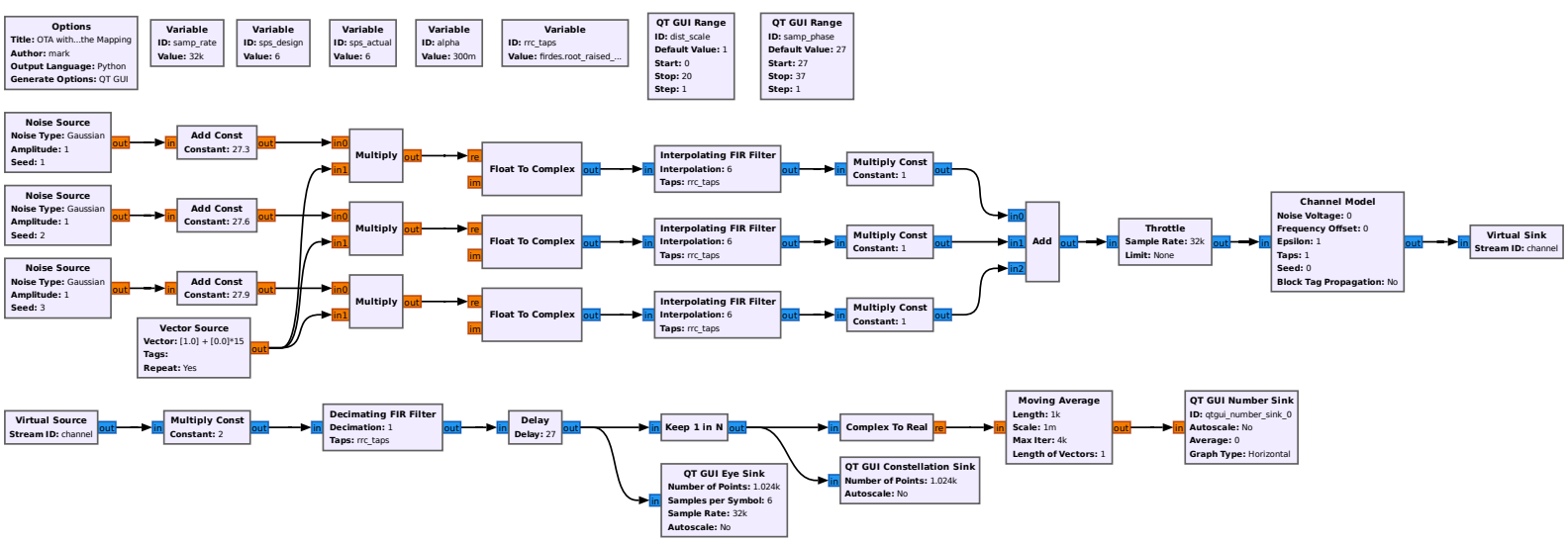


Figure 10: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

the introduction of timing offsets or Faster-Than-Nyquist signaling, the MAC superposition principle flawlessly computes the arithmetic average of the agricultural sensors.

A critical implementation detail warranting emphasis concerns the optimal sampling phase selection. Empirical observation from the GNU Radio receiver indicates that maximum data recovery is achieved at a sampling phase offset of 33 samples out of the 96-sample cycle ([1]). This phase offset reflects the group delay inherent to the polyphase interpolating and decimating FIR filter implementations, necessitating careful calibration for optimal demodulation performance.

[1]The fundamental cycle spans 16 symbols; with 6 samples per symbol, the total cycle length is $16 \times 6 = 96$ samples.

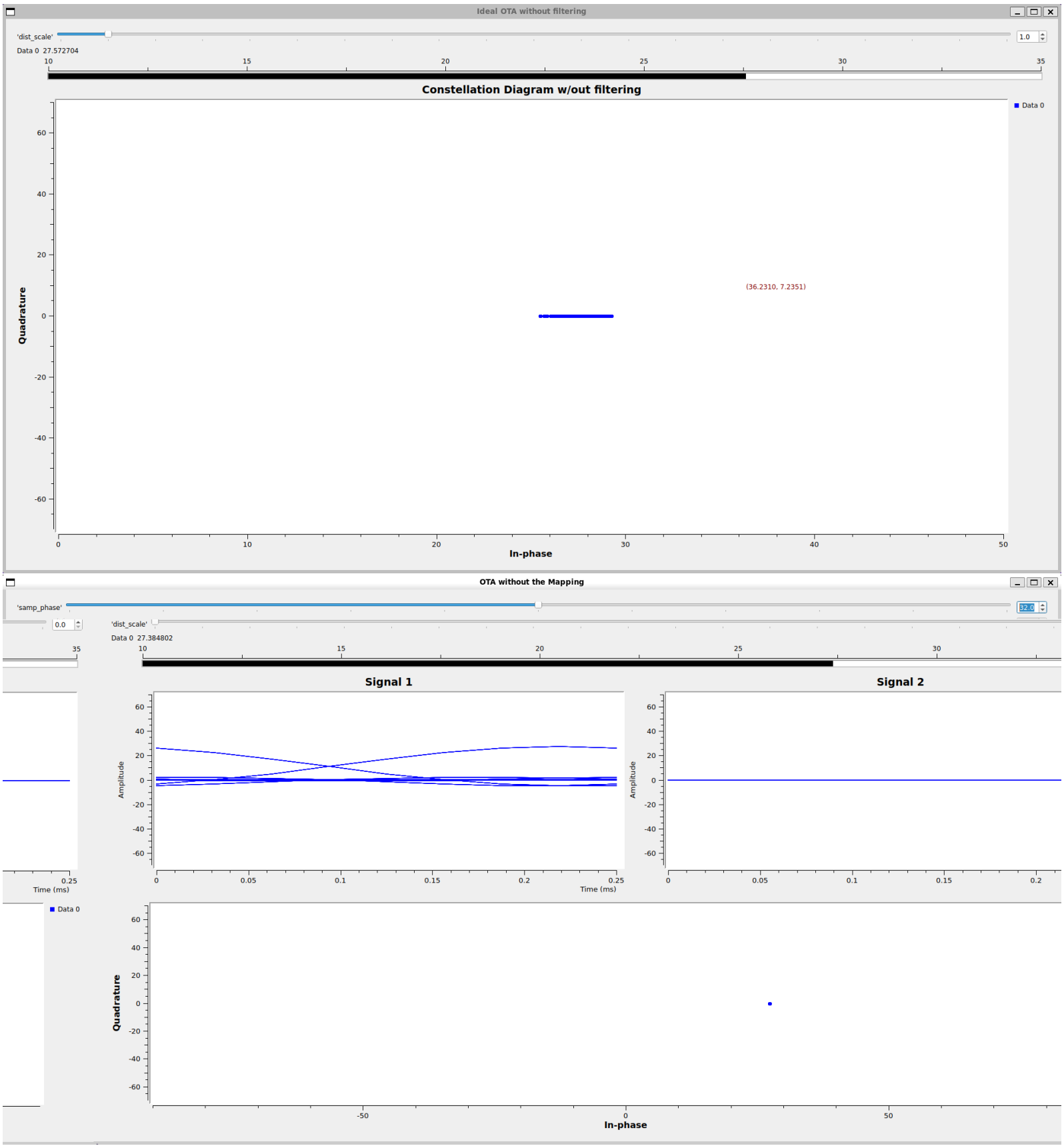


Figure 11: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

References

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